Double Propagation Stereo Matching for Urban 3-D Reconstruction From Satellite Imagery

Li Zhao, Yongmei Liu, Chaoguang Men, and Yubo Men

***Abstract*— Stereo matching is one of the popular methods to acquire the depth maps in 3-D reconstruction due to its high accuracy and completeness. However, traditional stereo matching methods are only suitable for general scenes or aerial images instead of urban scenes from large-scale satellite imageries. This article presents a novel double propagation stereo matching (DPSM) for urban 3-D reconstruction from satellite images. To make full use of the geometrical properties of man-made buildings, stereo-rectified satellite images, as input, are initialized to multiple superpixels with geometric models. Then, three similarity metrics are combined to calculate the weighted matching cost. A novel double propagation optimization is developed to optimize iteratively the weighted matching cost under the constraints of region boundaries and geometric models, and the disparity maps can be obtained by minimizing the energy function. Consequently, the corresponding dense 3-D point clouds and digital surface models (DSMs) are calculated by triangulation. Qualitative and quantitative experiments on stereo images captured by Pleiades and WorldView-2 satellites show that the proposed algorithm outperforms the most state-of-the-art stereo matching algorithms in terms of preserving depth discontinuous areas and restoring occlusions.**

***Index Terms*— 3-D reconstruction, double propagation, satellite images, stereo matching, urban scenes.**

I. Introduction

DURING the last decade, automatic 3-D reconstruction of urban scenes has been a research hotspot in the field of photogrammetry and computer vision, since that 3-D information plays a crucial role in urban intelligent traffic management, urban planning, and disaster management. Some scholars have proposed methods for automatic modeling of urban scenes on aerial images [1] and airborne Light Detection and Ranging (LiDAR) point clouds [2]. However, it is difficult for 3-D modeling from aerial imagery and LiDAR point clouds to expand to a wider radiating area due to various restrictions (e.g., traffic control, airline authorization, and equipment cost). Conversely, satellite imageries have overwhelming superiorities in terms of data cost and coverage range compared with aerial imageries. Furthermore, with the development of modern remote sensing technology, satellite sensors have been able to obtain images with submeter ground sampling distance (GSD) as well as high-overlapping stereo image pairs, which have been investigated and verified the availability and reliability in stereo matching on satellite images [3].

Recently, a 3-D reconstruction based on multistereo matching is of great interest as it is able to provide redundant information for generating high-quality 3-D products. However, two-view 3-D reconstruction is still of particular interest to intensively study because multistereo 3-D reconstruction is a direct extension of two-view stereo matching. In the existing two-view 3-D reconstruction scheme, 3-D information of scenes can be rebuilt based on disparity maps through triangulation and a series of processing steps. However, there appears to be significant room for obtaining accurate disparity maps about urban scenes of satellite images, potential problems are summarized as follows:

1) In contrast to frame cameras, linear array pushbroom sensors are used to acquire earth observation images during short acquisition time, resulting in different epipolar geometry which shows that the epipolar curves of the pushbroom cameras are not straight anymore. Therefore, the conjugate relationship of the nucleus is lost, which always causes a serious error during stereo rectification [4].
2) The baselines of images captured by satellites under the condition of high-speed motion are longer than aerial photographs. Therefore, there are plenty of inclined buildings on the satellite imageries, causing serious visual differences as well as numerous occlusions between satellite stereo image pairs.
3) More complex objects exist on the satellite images in urban scenes than natural scenes. Such as the disparity faults between roofs and facades as well as between roofs and grounds, the corners of buildings, the weak and similar textures on the flat ground and building roofs, and the different facades exhibited by the buildings from different perspectives all have put forward higher requirements for stereo matching algorithms [5].

To overcome the above limitations and obtain high-quality disparity maps within affordable time frames and computational resources, we propose the double propagation stereo matching (DPSM) for urban 3-D reconstruction from satellite images, and the results of reconstruction are shown in Fig. 1. In the proposed method, the disparity maps are calculated

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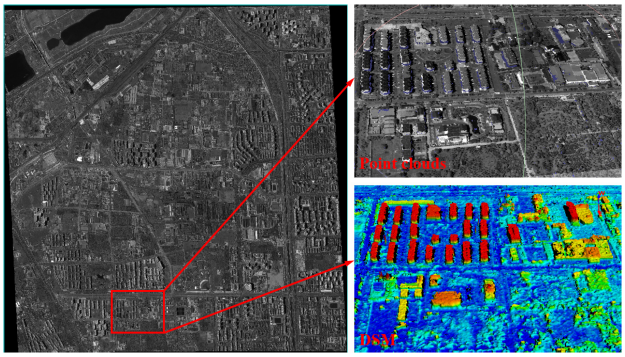

Fig. 1. Overview of 3-D recovery from two-view satellite images.

based on matching cost optimized by the double propagation, which consists of the external propagation and the internal propagation. More concretely, given that surface space is capable of providing valuable clues for region segmentation and optimizing matching cost in-depth discontinuous areas and oblique surfaces (e.g., the inclined building tops and the sides of buildings), the external propagation that takes the surface regions as primaries iteratively fuses regions and updates regional geometric models for promoting the optimization of matching cost. Additionally, the internal propagation that takes pixels as primaries carries out the penalty propagation for optimizing matching cost pixel by pixel under the restriction of region boundaries from the external propagation. Therefore, the accurate disparity maps can be worked out by minimizing the energy function based on the optimized matching cost. In general, the proposed algorithm consists of the initialization of regional models, calculation of joint weighted matching cost and double propagation. *1) Initialization of regional models:* Superpixel segmentation and downsampling matching are first executed on stereo-rectified satellite images, yielding superpixel regions and uniformly distributed matching points. Based on these matching points, spatial models corresponding to superpixels are obtained by model fitting. *2) Calculation of joint weighted matching cost:* The weighted sum of patch cost, Census cost [6] and Pyramid Histogram Of Visual Words (PHOW) cost [7] is employed to acquire the matching cost. *3) Double propagation:* The cost of depth discontinuous, inclined surfaces and occlusive areas can be optimized by double propagation which consists of the external propagation and the internal propagation.

In this article, the digital surface models (DSMs) and 3-D point clouds that are calculated based on the disparity maps from traditional and state-of-the-art stereo matching algorithms are used to evaluate the proposed algorithm. Moreover, the qualitative and quantitative comparison experiments have shown that DPSM not only can calculate accurate disparity values of depth discontinuous areas but also resist the interferences of inclined buildings and occlusions. The main contributions of our work can be summarized as follows:

1) A new downsampling matching method based on PHOW features has been presented to acquire uniformly distributed matching points in the step of initialization of regional models. The proposed downsampling matching method first obtains uniformly distributed candidate points through downsampling, followed by building matching relationships by using PHOW features. Obviously, it is beneficial for adjustment calculation, stereo correction, and structure from motion (SfM) because such tasks usually require nonconcentrated matching points. In particular, it is still a challenging task to obtain uniformly distributed matching points on satellite images with changeable illumination.
2) Unlike traditional stereo matching based on segmentation (e.g., Ref. [8] and Ref. [9]), double propagation optimization is adopted in our method which is robust to depth discontinuous areas, inclined surfaces, and occlusions. Stereo images are first segmented into multiple regions, in traditional segmentation-based stereo matching algorithms, then edge features are used to improve matching cost. Nevertheless, new errors may be introduced due to inaccurate area edges. Conversely, wrong area edges can be filtered out dynamically and the true edges of depth discontinuous areas can be retained during the external propagation, thus the matching cost is able to be optimized under the constraints of true edges during the internal propagation. Furthermore, the proposed optimization of double propagation does not rely on the calculation methods of matching cost, in essence, which is an optimization method of matching cost and can be applied to optimizing matching cost of other stereo methods (e.g., the matching cost of Census in Ref. [10], the matching cost of Mutual Information in Ref. [11] and the matching cost of support window in Ref. [12]–[14]) for disparity refinement as well.

II. Related Work

Numerous methods about 3-D reconstruction of urban scenes have been proposed, most of which focus on aerial images [8], [15] and modeling of point clouds obtained by LiDAR [2], [16]. Due to a low cost of acquisition and equipment, satellite-image-derived 3-D reconstruction is more suitable for large-scale urban scenes. However, there are still many problems in applying image-derived 3-D reconstruction to satellite images.

First of all, the original satellite images are failed to be directly used to stereo matching, since satellite images from pushbroom sensors have nonstraight epipolar curves, the conjugation between epipolar curves is destroyed as well [4]. Several methods for stereo rectification of pushbroom images have been proposed [17], [18], and it has become a very popular practice to make use of Rational Polynomial Coefficients (RPC) model instead of the complicated linear-array sensor model, such that the modelers do not need to create different types of camera models according to parameter protocols from different satellite sensors and platforms [19]. An approximation scheme based on RPC using an affine camera model, proposed by Franchis *et al.* [20], is the most popular solution for stereo rectification. In our 3-D recovery pipeline, the RPC pushbroom rectification method proposed by

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Franchis *et al*. is employed to obtain stereo-rectified satellite images which act as the inputs of DPSM.

Unlike LiDAR that generates independent 3-D measurements of the ground objects per light beam, image-derived 3-D reconstruction is largely dependent on disparity results, thus stereo matching has always been a key step in 3-D reconstruction. To yield better matching results in texture-less areas, the widely used methods often introduce more matching features in support windows [12], [13], [21]. Yet there is an implicit assumption in this procedure that all pixels in the support window have a constant parallax, which may cause extra buffers of edges and lead to extended edges in the final disparity maps [e.g., the red boxes in Fig. 15(c) and (e)]. An effective way to solve this problem is local filtering which is able to achieve *local space smoothing* in the labeled space, and higher-quality results can be yield in spite of this conceptual flaw [14] [e.g., the red boxes in Fig. 15(g)].

There is an urgent need for edge constraints, and ongoing efforts tend to incorporate the image edges to yield better matching results. Duan and Lafarge [9] proposed a large-scale urban reconstruction method for satellite images, in which satellite images are divided into multiple polygonal regions containing semantic information. Therefore, semantics features and geometric features inside polygon areas can be utilized to estimate the parallaxes. Similarly, satellite images are divided into a specified number of blocks by superpixel segmentation in the method proposed by Hou *et al*. [8]. Additionally, edge regions are extracted as matching features to avoid incorporating depth discontinuity in other works [22], [23]. Unfortunately, it is not enough to identify depth faults solely based on edge features, for that region boundaries are difficult to attach real disparity faults. New parallax errors are easily introduced due to the incorrect positioning of the depth discontinuous areas [e.g., Fig. 15(f)]. To tackle this issue, we propose the external propagation based on belief propagation (BP) algorithm in which the spatial model of each superpixel, as geometric primitive, can be propagated among different superpixels. Even if there are a few inaccurate region boundaries or spatial models, they will be optimized iteratively in the next round of propagation (See Section III-C for details).

In addition, even though many stereo matching methods based on the deep neural network have top rankings in stereo benchmarks, which are failed to be used on satellite images due to lack of ground truth disparity results of satellite images for training [24]–[26]. Instead, semi global matching (SGM), as a traditional method, has been widely used in the extraction of DSM on satellite images for its ideal accuracy and computational complexity [11]. Sajid [18], Tatar *et al*. [27], and Mandanici *et al*. [28] directly applied SGM to the acquisition of disparity maps on stereo-rectified satellite images without any modification. To deal with the noises and ambiguities of similarity metrics in texture-less areas, existing modified methods based on SGM normally tend to expand the dimension of feature space [29]. Li *et al*. [10] applied the modified SGM algorithm on the worldview-2 satellite images by replacing mutual information entropy, in original SGM, with Census transform. In essence, Census represents local characteristics between the center pixel and the surrounding pixels. Although Census transformation has been proven to be robust to the changes of illumination, the similarity degree about the gray relationship often leads to the ambiguities between the target pixel and multiple candidate pixels [e.g., Fig. 15(e)]. Consequently, to make full use of the property of Census, we utilize Census cost as one term of the joint weighted matching cost. The improved cost is not only robust to illumination variation but also able to avoid the loss of uniqueness of matching (See Section III-B for details).

In order to optimize matching cost, 2-D smoothness constraint is transformed as the average of multiple 1-D paths (usually with 8 cardinal directions) in SGM. However, star-shaped streaking artifacts often appear in-depth discontinuous areas, because 8-direction path aggregation uses each pixel as the center. To deal with this problem, Facciolo *et al*. [30] proposed a significantly more global matching (MGM) stereo matching method which extended the 1-D cost aggregation of SGM to the quadrant cost aggregation. Moreover, there is a post-processing approach presented by Qin *et al*. [31] who took the matched straight lines as the units to refine parallaxes achieved from SGM. Unfortunately, the inaccurate modifications might submit to the wrong straight lines since that the straight lines were hypothesized to represent depth discontinuity edges. Dynamically optimized region edges, just like matching lines, are used to refrain from the influence of erroneous priors in the proposed method. Inspired by 1-D path aggregation of SGM and local space smoothing, the scanning-line penalty propagation based on local regions is developed in this article, namely internal propagation, for yielding accurate matching cost in depth discontinuous areas (See Section III-C for details).

III. DPSM Algorithm

In general, the proposed approach consists of three steps: initialization of regional models, calculation of joint weighted matching cost, and double propagation. First, the left/right view of stereo-rectified satellite images is divided into multiple superpixels, and the regional model of each superpixel can be initialized by means of random sample consensus (RANSAC) based on downsampling matching points in the step of initialization of regional models. Then, as the initial matching cost, the joint weighted matching cost is calculated to yield similarity distances between corresponding pixels. Finally, the proposed double propagation is used to iteratively optimize the initial matching cost and acquire the precise disparity maps. An overview of the pipeline is given in Fig. 2.

A. Initialization of Regional Models

The purpose of the initialization of regional models is to achieve region division and geometric model corresponding each region. Thus, superpixel segmentation, downsampling matching based on PHOW features and model fitting are included in this section.

1) Superpixel Segmentation: A superpixel is an image block composed of adjacent pixels with similar texture, color, and brightness characteristics. Therefore, pixels in one superpixel may have the same geometric features as well as similar parallaxes. For the sake of concentrating on acquiring region

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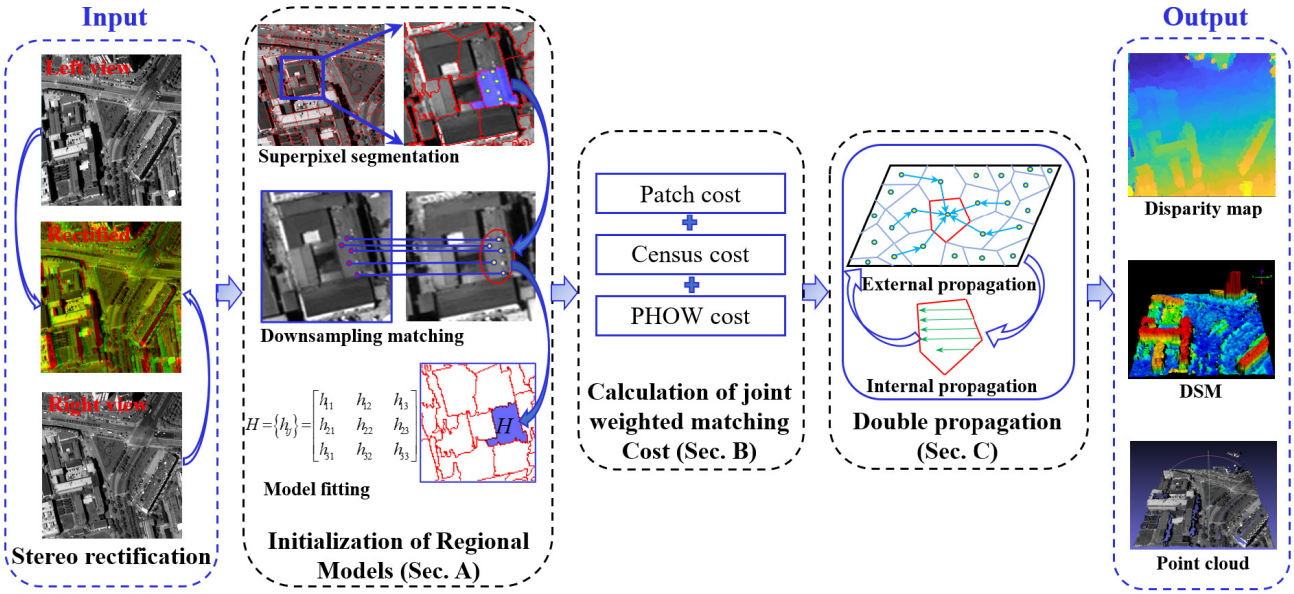

Fig. 2. Outline of our approach.

division of different geometric entities (over-segmentation is permitted), the simple linear iterative clustering (SLIC) superpixel algorithm is employed to segment satellite images [32]. Compared with other superpixel algorithms such as entropy rate superpixel (ERS) [33] and superpixels extracted via energy-driven sampling (SEEDS) [34], SLIC is the best choice in our task. For that superpixel boundaries are crucial to the location of depth discontinuities areas, smooth and complete edges are essential in the proposed method. ERS achieves segmentation by the graph topology under the guidance of maximizing the objective function which consisted of the entropy rate of a random walk and a balancing term. The superpixels produced by ERS are irregular in regions with complicated texture because the balancing term only limits size of superpixels. SEEDS introduced an energy function that encouraged color to get uniform distribution and stimulated boundary shapes to be smooth. However, the size of superpixels is affected more seriously when images have very irregular shape content. Conversely, the smooth and regular area edges can be obtained by adjusting the regularization item and the area size item in SLIC. In urban scenes, smooth and regular superpixels can attach to the real edges of buildings with geometric shapes and be robust to uneven light and noise, thus they will provide accurate edge constraints for optimizing parallaxes of depth discontinuous areas in the internal propagation. Specifically, in the proposed method, the area size item of SLIC is set by referring to the SLIC segmentation accuracy on Berkeley data set (BSD), and the regularization item of SLIC can be set through repeated experiments.

In addition, the computational complexity of SLIC, ERS and SEEDS are $O(n)$, $O(nN^2 \lg N)$ and $O(N \lg N)$, respectively. Thus, SLIC is the fastest one among the three superpixel segmentation algorithms and accurate segmentation results can be quickly achieved through SLIC in the initialization of regional models. Benefit from the good dependence of real object boundaries, the superpixel edges are able to *locate* disparity faults accurately and ensure the aggregation of pixels with similar parallaxes.

As a result, a stereo-rectified satellite image is segmented into n superpixels $\{S_i\}_{i=1}^{n}$ by SLIC, where i represents the superpixel label.

2) Downsampling Matching Based on PHOW: Precise model fitting relies on sufficient matching points inside each superpixel. Unfortunately, in contrast to this target, it is formidable for feature matching to get enough matching points in each superpixel. Therefore, we propose the downsampling matching based on the PHOW which was first used to image classification in Ref. [7]. In order to improve the robustness of PHOW features in different environments, the PHOW descriptor of a pixel is achieved by calculating the sum of scale-invariant feature transform (SIFT) descriptors with different radii [35]. To be specific, many objects in urban scenes, such as building facades, occupy very few pixels on the satellite images, due to different shooting angles of the satellite sensors. For this situation, it is easy for the PHOW feature with a large radius to ignore local subtle details. Conversely, it is also tough for the PHOW feature with a small radius to obtain enough features for matching in flat areas. Using the superposition of multiple SIFT features with different radii is able to overcome problems above. At the same time, considering that the Gaussian filter is used to construct a multiscale pyramid in the process of extracting SIFT feature, an effective SIFT feature will not be extracted if the selected radius is too small. Consequently, the minimum radius is determined to be 4 pixels, and other radii are increased gradually in multiples of the minimum radius, and the radii of the circle support areas are set to be 4, 8, and 12 pixels in the proposed method, the examples of two different areas are shown in Fig. 3.

The PHOW descriptor can be acquired by the following equation:

$$\mathrm{PHOW}(x, y) = \mathrm{SIFT}_4(x, y) + \mathrm{SIFT}_8(x, y) + \mathrm{SIFT}_{12}(x, y). \tag{1}$$

Accordingly, uniformly distributed matching points with the specified number can be received in each superpixel of satellite

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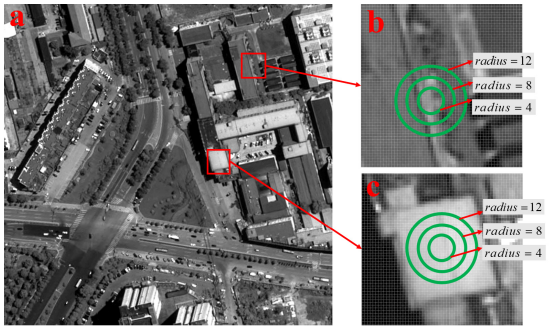

Fig. 3. Examples of PHOW features in different areas. (a) Satellite image. (b) and (c) Enlarged subregions corresponding to the red boxes in (a). In (b) and (c), the green circles represent the boundaries of feature areas with three radii.

Algorithm 1 Downsampling Matching Based on PHOW

Input: Stereo-rectified left image I_L, stereo-rectified right image I_R, downsampling factor T_d, minimum disparity value min_disp, maximum disparity value max_disp

Output: The set of downsampling matching points $\mathcal{P}_{down}^{l\leftrightarrow r}$

for *each pixel* p_L *step* T_d, $p_L \in I_L$ **do**
  Extract PHOW feature $PHOW(p_L)$ at p_L;
for *each pixel* p_R *step* T_d, $p_R \in I_R$ **do**
  Extract PHOW feature $PHOW(p_R)$ at p_R;
Get indexes of matching points $\mathcal{P}_{down}^{l\leftrightarrow r}$ between feature point sets $\{PHOW(p_L^i)\}_{i=1}^n$ and $\{PHOW(p_R^i)\}_{i=1}^n$ within $[min_disp, max_disp]$ by NNDR strategy (e.g., 2);
return $\mathcal{P}_{down}^{l\leftrightarrow r}$;

images (e.g., Fig. 9). Note that unlike traditional feature matching, a few false matches are allowed in the proposed downsampling matching, for that false matching points would be filtered out in the step of model fitting.

Subsequently, the proposed downsampling matching method can get the set of descriptors $\mathcal{P}_{\text{down}}$ at T_d pixel intervals, then the set of matching points $\mathcal{P}_{\text{down}}^{l\leftrightarrow r}$ is calculated by the nearest neighbor distance ratio (NNDR) strategy with threshold Thr, as shown in the following formula:

$$\frac{\text{Min}\left(\left\{\left\|\text{PHOW}(x_l, y_l) - \text{PHOW}\left(x_r^i, y_r^i\right)\right\|_2\right\}_{i=1}^n\right)}{\text{Sec}\left(\left\{\left\|\text{PHOW}(x_l, y_l) - \text{PHOW}\left(x_r^i, y_r^i\right)\right\|_2\right\}_{\text{i=1}}^n\right)} < \text{Thr} \quad (2)$$

where $\text{Min}(\cdot)$ and $\text{Sec}(\cdot)$ are the function of the minimum and the second minimum, respectively. $\text{PHOW}(x_l, y_l)$ represents the PHOW feature at point (x_l, y_l) in left view. $\text{PHOW}(x_r^i, y_r^i)$ represents the PHOW feature at the i_{th} point in right view. Besides, in order to improve the efficiency of downsampling matching, the search range of matching points can be restricted between the minimum disparity value and the maximum disparity value. The proposed downsampling matching algorithm is summarized as Algorithm 1.

3) Model Fitting: Generally, the geometrical models that represent the variation trend of parallaxes inside superpixels are able to provide spatial constraints for calculating parallaxes

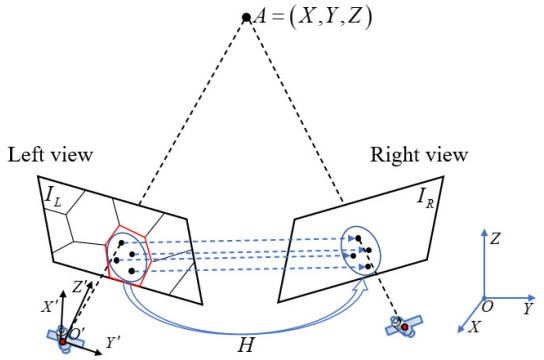

Fig. 4. Homography relationship between superpixels on binocular satellite images. Where, $O - XYZ$ is the world coordinate system, $O' - X'Y'Z'$ is the camera coordinate system, I_L and I_R are left view and right view, respectively. The red polygonal region on I_L is a superpixel in which the blue dotted arrows indicate the pairs of downsampling matching points, H is a homography matrix corresponding to this superpixel.

of oblique surfaces in urban scenes. In the proposed method, homography models between the corresponding areas are fit by multi-RANSAC algorithm proposed by Zuliani *et al.* [36] under the condition of sufficient support matching points. RANSAC will consider that the spatial feature of the current superpixel is 0, if the number of noncollinear matching points is insufficient inside a superpixel. Such that only color feature participates in similarity calculation between adjacent superpixels during the external propagation. The probability of this situation is very low because the downsampling matching based on PHOW has been proposed to provide sufficient matching points for fitting homography models. The schematic of the homography relationship between superpixel pairs is shown in Fig. 4. In addition, model fitting is not only performed in the initialization of the regional models, but also utilized to update the regional models in the external propagation.

Based on this homography matrix, the point $p_1 = [x_1, y_1, w_1]^T$ can be converted into a new point $p_2 = [x_2, y_2, w_2]^T$ through transformation formula $H p_1 = p_2$, where H is a 3×3 homography matrix, as follows:

$$H = \{h_{ij}\} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}. \quad (3)$$

To normalize, let the bottom-right-corner element $h_{33} = 1$. Thus, a homography matrix has eight free elements $h_{11}, h_{12}, h_{13}, h_{21}, h_{22}, h_{23}, h_{31}, h_{32}$ left. The detailed derivation process is as follows.

Assume that one pair of matching points on the two views are $(x_1, y_1) \in \mathcal{P}_{\text{down}}$ and $(x_2, y_2) \in \mathcal{P}'_{\text{down}}$, their homogeneous coordinates are $p_1 = [x_1, y_1, 1]^T$ and $p_2 = [x_2, y_2, 1]^T$. Substituting them into the transformation formula $H p_1 = p_2$, the following results can be obtained:

$$\begin{cases} x_2 = \dfrac{x_1 h_{11} + y_1 h_{12} + h_{13}}{x_1 h_{31} + y_1 h_{32} + 1} \\ y_2 = \dfrac{x_1 h_{21} + y_1 h_{22} + h_{23}}{x_1 h_{31} + y_1 h_{32} + 1}. \end{cases} \quad (4)$$

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Then, we can get

$$\begin{cases} x_1h_{11} + y_1h_{12} + h_{13} - x_1x_2h_{31} - x_2y_1h_{32} = x_2 \\ x_1h_{21} + y_1h_{22} + h_{23} - x_1y_2h_{31} - y_1y_2h_{32} = y_2. \end{cases} \tag{5}$$

The equation above can be converted into the form of $Au = v$

$$A = \begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -x_1x_2 & -x_2y_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -x_1y_2 & -y_1y_2 \end{bmatrix} \tag{6}$$

$$u = [h_{11} \quad h_{12} \quad h_{13} \quad h_{21} \quad h_{22} \quad h_{23} \quad h_{31} \quad h_{32}]^T \tag{7}$$

$$v = [x_2, y_2]^T. \tag{8}$$

This is due to the fact that one pair of matching points can provide two equations from (5). Thus, an eight-row simultaneous equation with a unique solution can be formed by four pairs of noncollinear matching points. Subsequently, the singular value decomposition (SVD) decomposition is utilized to solve homography matrix H.

B. Calculation of Joint Weighted Matching Cost

Taking account of the influences of building corners, weak textures, and illumination variation, we combine three kinds of similarity measures to calculate the matching cost between corresponding points on satellite images. The three matching costs employed in the proposed algorithm includes patch cost, Census cost [6] and PHOW cost [7], and the formula of joint weighted matching cost is as follows:

$$\text{COST} = \alpha \cdot \text{COST}_P + \beta \cdot \text{COST}_C + \gamma \cdot \text{COST}_{\text{PH}} \tag{9}$$

where α, β and γ indicate weights of different costs and the values are 0.3, 0.2 and 0.5, respectively. Additionally, the effectiveness of these parameters is evaluated in *IV Experiments C. The Evaluation of Three Matching Costs*. The COST_P, COST_C and COST_{PH} represent patch cost, Census cost and PHOW cost, respectively. The three matching costs are employed according to the characteristics of the urban satellite imageries. Patch cost is robust to weak and similar textures distributed on the surfaces of the buildings in urban scenes, and Census cost can refrain from illumination variation of satellite images. In addition, PHOW cost is utilized to yield parallaxes at the corners of man-made buildings with obvious geometric elements. The calculation of the three matching costs is shown below.

1) Patch Cost: Given that gray and gradient inside sliding windows are robust to weak and similar textures on urban satellite imageries, the weighted sum of gray distance and gradient distance is chosen as the patch cost, as follows:

$$\text{COST}_P = \frac{1}{{S_W}^2} \sum_{q \in W_p} W(p,q)\rho(q,q') \tag{10}$$

where W_p represents a square window centered on pixel p and S_W is the size of this window. The weight function $W(p,q)$ measures the perceived importance of every pixel inside a sliding window, which is designed to overcome the obstacles from object distortion in urban scenes. More directly, the weight should decrease with the increase of color difference as well as spatial distance. Thereby, the gray difference and spatial distance between the center pixel p and the other pixels are employed, as follows:

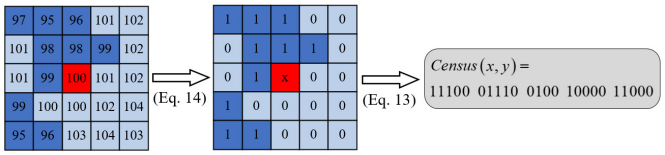

Fig. 5. Analysis of a Census transformation. The size of this window is 5×5, the red box is the center pixel $p(x, y)$ of this window and its gray value is 100, the dark blue boxes are pixels whose gray values are less than 100, on the contrary, the light blue boxes are pixels whose gray values are 100 or more. When Census transformation is performed in this window, the bit values of all noncentral pixels inside this window are obtained by (14), and the result of Census transformation centered on pixel $p(x, y)$ is obtained by (13).

$$W(p,q) = e^{-\frac{\|I_p - I_q\|_1}{\kappa} - \frac{\|C_p - C_q\|_2}{S_W}} \tag{11}$$

where κ is utilized to balance gray difference and Euclidean distance. $\|I_p - I_q\|_1$ represents the $L1$ distance between the gray values of pixel p and q, $\|C_p - C_q\|_2$ represents the $L2$ distance between the coordinates of pixel p and q.

In (10), the function $\rho(q, q')$ calculates the pixel difference between different views

$$\rho(q,q') = (1-\omega) \cdot \min(\|I_q - I_{q'}\|_1, \tau_{\text{gray}}) + \omega \cdot \min(\|\nabla I_q - \nabla I_{q'}\|_1, \tau_{\text{grad}}) \tag{12}$$

where ω is the weight between gray difference and gradient difference. $\|I_q - I_{q'}\|_1$ represents the $L1$ distance of gray value between pixel q and q'. $\|\nabla I_q - \nabla I_{q'}\|_1$ represents the $L1$ distance of gradient value between pixel q and q'. Pixel q' is the matching point of q. To make COST_P be robust to occluded areas, the parameters τ_{gray} and τ_{grad} are proposed to represent the cutoff values of the gray difference and gradient difference, respectively, which has been utilized in Ref. [37].

2) Census Cost: The calculation of Census cost is as described below

$$Cs(x,y) = \bigotimes_{\eta=\lfloor -\frac{m}{2} \rfloor}^{\eta=\lfloor \frac{m}{2} \rfloor} \bigotimes_{\xi=\lfloor -\frac{n}{2} \rfloor}^{\xi=\lfloor \frac{n}{2} \rfloor} \delta(I_p(x,y), I_p(x+\eta, y+\xi)) \tag{13}$$

$$\delta(x_1, x_2) = \begin{cases} 1, & x_2 < x_1 \\ 0, & x_2 \geq x_1 \end{cases} \tag{14}$$

where $I_p(\cdot)$ is the gray value of pixel p, and $m \times n$ is the size of square window centered on pixel $p(x, y)$. The $\otimes$ represents the back-and-forth join operation based on the results of operation $\delta(\cdot)$. For simplicity, an example of Census transformation is shown in Fig. 5.

Hamming distance is usually exploited to calculated Census cost between two lists of bit strings. Formally, let two lists of bit strings of pixels from two views are $Cs(x_l, y_l)$ and $Cs(x_r, y_r)$, the Census cost between the two pixels is

$$\text{COST}_C = \text{Dist}_{\text{HAMMING}}(Cs(x_l, y_l), Cs(x_r, y_r)) \tag{15}$$

where $\text{Dist}_{\text{HAMMING}}(\text{str1}, \text{str2})$ represents the number of different bit values at corresponding positions between $str1$ and

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$str2$. More straightforward, the smaller Hamming distance is, the more similar two pixels are.

3) PHOW Cost: The PHOW was first proposed for image classification by Bosch *et al.* [7]. Owing to the excellent accuracy of SIFT descriptors, the PHOW descriptors are robust to the building corners and weak textures in urban scenes. PHOW is treated as a kind of special *matching cost* to calculate the PHOW cost at each pixel in our algorithm. In order to improve the calculation efficiency, we simplify the extraction of PHOW descriptor by a single SIFT descriptor with radius 8, as follows:

$$\mathrm{PHOW}(x, y) = \mathrm{SIFT}_8(x, y). \tag{16}$$

The PHOW cost is obtained by calculating the Euclidean distance between the two PHOW descriptors of corresponding points, as follows:

$$\mathrm{COST}_{\mathrm{PH}} = \|\mathrm{PHOW}(x_l, y_l) - \mathrm{PHOW}(x_r, y_r)\|_2. \tag{17}$$

C. Double Propagation

The double propagation model consists of external propagation and internal propagation. Essentially, the external propagation is a kind of regional information propagation among superpixels based on BP [38], which is utilized to iteratively fuse superpixel regions and update their corresponding spatial models. In addition, the internal propagation is a kind of pixel-wise penalty propagation, which can optimize the pixel-wise matching cost pixel by pixel. In the end, a precise disparity map is able to be yielded by minimizing the energy function.

1) External Propagation: The basic assumption of the external propagation is that pixels with similar color and spatial characteristics may share similar parallaxes or a similar variation of parallaxes, thus the problem of cost optimization is transformed into region division. Under this mechanism, the desired superpixels with boundaries aligned with disparity faults and corresponding spatial models can be optimized by updating iteratively superpixels and their corresponding models.

More narrowly, a probabilistic graph model is constructed based on superpixel results. Superpixels are regarded as the nodes and adjacent relationship between superpixels are treated as the edges (e.g., Fig. 6). To measure the degree of similarity between adjacent superpixels (i.e., nodes in the graph model), the max-product BP is used to calculate the Maximum A Posterior (MAP) estimator in external propagation.

During each iteration, each node uses the messages it has received from neighbor nodes in previous iteration to calculate messages [39]. However, the problem arises when the *loop structure* occurs in this model which might lead the message propagation to be stuck in an infinite loop. To solve the propagation cycles, at each iteration, a *target node* which acts as a receiver and does not propagate message out is selected out according to the size of the corresponding superpixel (e.g., node i of graph G in Fig. 6). After the *target node* is merged into a new node, the new node will continue to propagate messages and another *target node* will be selected out again. The graph structure of external propagation is shown in Fig. 6.

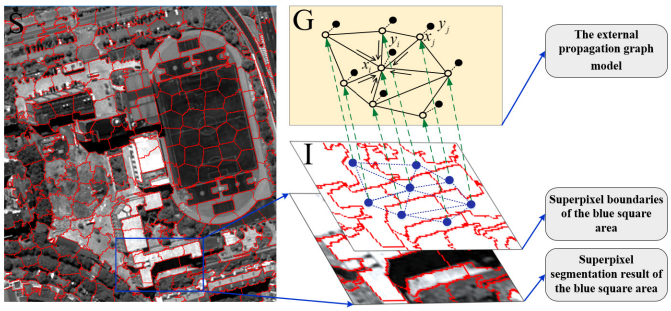

Fig. 6. External propagation. The left image S is a superpixel segmentation result from a satellite image. I is superpixel boundaries of the blue box area in S, and G is external propagation graph model corresponding to the I. In G, the white nodes are implicit nodes and the black nodes are explicit nodes. The dotted lines represent the likelihood functions (e.g., $\phi_i(x_i, y_i)$ is simplified as $\phi_i(x_i)$) existing between implicit and explicit nodes. The solid lines indicate pairs of potential function (e.g., ψ_{ij} and ψ_{ji}). The arrows indicate directions of message propagation between nodes. Furthermore, node i is the target node in current iteration.

The belief of node i is the product of all messages which are sent to node i, as follows:

$$b_i(x_i) = \lambda \phi_i(x_i) \prod_{j \in \mathcal{N}(i)} m_{j \to i}(x_i) \tag{18}$$

where λ is a normalization constant, $\phi_i(x_i)$ is likelihood function between implicit nodes and explicit nodes, $\mathcal{N}(i)$ represents a set of nodes adjacent to node i, and $m_{j \to i}(x_i)$ represents message that node i has received from node j. For concreteness, the calculation of $\phi_i(x_i)$ is as follows:

$$\phi_i(x_i) = \prod_{p \in \mathcal{P}} \frac{1}{\sqrt{2\pi}\delta_i} \exp\left(-\frac{(f_p - \mu_i)^2}{2\delta_i^2}\right) \tag{19}$$

where $\mathcal{P}$ is a set of pixels inside superpixel i, p is the pixel in $\mathcal{P}$ and f_p is the image feature of the p_{th} pixel. δ_i and μ_i are standard deviation and mean value of pixels inside superpixel i.

The message of current node is related to the product of all the messages come from the previous one, so the update rules of messages are as follows:

$$m^t_{j \to i}(x_i) = \sum_{x_i} \phi_i(x_i) \psi_{j \to i}(x_i, x_j) \prod_{k \in \mathcal{N}(j)/i} m^{t-1}_{k \to j}(x_j) \tag{20}$$

where $m^t_{j \to i}(x_i)$ represents the message transmitted from node j to neighboring node i at the t_{th} iteration. $\mathcal{N}(j)/i$ represents the neighborhood of node j excluding node i, and $\psi_{j \to i}(\cdot)$ represents potential function between node j and i, as follows:

$$\psi_{j \to i}(x_i, x_j) = \theta \cdot \mathrm{DisG}(x_i, x_j) + (1 - \theta) \cdot \mathrm{DisH}(x_i, x_j) \tag{21}$$

where θ is utilized to balance the gray distance $\mathrm{DisG}(\cdot)$ and spatial distance $\mathrm{DisH}(\cdot)$. $\mathrm{DisG}(x_i, x_j)$ is formulated as follows:

$$\mathrm{DisG}(x_i, x_j) = \left\| \sum_{p \in \mathcal{P}_{x_i}} I_p \Big/ \mathrm{Num}(p) - \sum_{p \in \mathcal{P}_{x_j}} I_p \Big/ \mathrm{Num}(p) \right\|_1 \tag{22}$$

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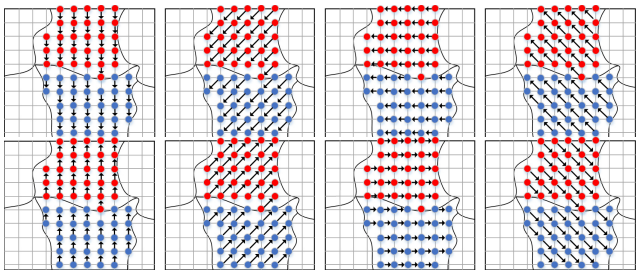

Fig. 7. Internal propagation. (a)–(h) Schematic diagrams of scanning-line penalty propagation with eight directions under the circumstance of two superpixels. The black curves are boundaries of superpixels, the red and blue points represent pixels located in different superpixels.

where $\mathcal{P}_{x_i}$ and $\mathcal{P}_{x_j}$ are sets of pixels inside superpixel i and superpixel j, respectively. I_p is the gray value of pixel p, and $\mathrm{Num}(p)$ is the number of pixels inside a superpixel.

$\mathrm{DisH}(x_i, x_j)$ is achieved by calculating the distance between downsampling matching points of a superpixel and the homography model of its adjacent superpixel. Furthermore, the purpose of homography-model distance is to quantify the spatial difference between superpixel i and j, thus the specific form of spatial difference is as follows:

$$\mathrm{DisH}(x_i, x_j) = \sum_{t \in \mathcal{T}_{x_i}} \| H_j \cdot l^t, l^{t'} \|_2 / \mathrm{Num}(t) + \sum_{t \in \mathcal{T}_{x_j}} \| H_i \cdot l^t, l^{t'} \|_2 / \mathrm{Num}(t) \quad (23)$$

where $\mathcal{T}_{x_i}$ and $\mathcal{T}_{x_j}$ indicate the sets of matching points inside superpixel i and j, respectively, which are subject to $\mathcal{T}_{x_i} \subseteq \mathcal{P}_{\mathrm{down}}, \mathcal{T}_{x_j} \subseteq \mathcal{P}_{\mathrm{down}}$. $\| H_j \cdot l^t, l^{t'} \|_2$ represents the Euclidean distance between the transformation of coordinate l^t using homography model H_j and corresponding matching point coordinate $l^{t'}$. Note that the matching point t and homography model H_j belong to different superpixels. $\| H_i \cdot l^t, l^{t'} \|_2$ has a similar meaning.

In the end, the label of max belief can be computed through the max-product BP, as follows:

$$L(j_{max}) = \arg\max_{j \in \mathcal{N}(i)} b(x_i). \quad (24)$$

2) Internal Propagation: The internal propagation is a scanning-line penalty propagation with region constraint based on SGM. Notice that unlike the 1-D line optimization of SGM or the quadrant cost aggregation of MGM, the proposed internal propagation optimizes cost under the constraints of region boundaries come from external propagation. Moreover, considering that disparity jumps often exist at the region boundaries, penalties imposed on region edges are enhanced during scanning-line penalty propagation. The internal propagation with eight directions employed in this article is shown in Fig. 7.

During the internal propagation, the matching cost of each pixel is punished along the paths of scanning lines under the constraints of superpixel boundaries. And in particular, the penalty function is formulated as follows:

$$V^s(D_p, D_q) = \begin{cases} 0, & \text{if } |D_p - D_q| = 0 \\ p_1, & \text{if } |D_p - D_q| = 1 \\ p_2, & \text{if } |D_p - D_q| = D_p \\ p_3, & \text{others} \end{cases} \quad (25)$$

where p_1 penalizes the case that disparity change is 1 (pixel), p_2 penalizes the case that pixels lie on disparity faults (the disparity value of pixel q in neighboring superpixel is 0 after binary processing, thus $|D_p - D_q| = D_p$), p_3 penalizes the case that the disparity change is greater than 1 (pixel), moreover p_1, p_2 and p_3 are subject to $p_2 > p_3 > p_1$. Comparing with standard SGM, the improved penalty function not only permits similar disparities or slightly varying disparities, but also permits tremendous disparity changes between adjacent superpixels where depth faults often appear. Even if there are wrong superpixel divisions, which will be integrated in the next round of propagation.

Under the constraints of superpixel boundaries, matching cost is optimized iteratively starting from one side to the other side of superpixel s, and the penalty propagation along path r is a limited iterative procedure

$$L_r^s(p, d) = C^s(p, d) + \min_{q \in \mathcal{N}^r(p)} \left(L_r^s(p - r, D_q) + V^s(D_p, D_q) \right) \quad (26)$$

where q is the previous pixel of p on path r, and $\mathcal{N}^r(p)$ is the neighborhoods of pixel p on path r inside superpixel s. The energy function of cost inside superpixel s is calculated by summing costs on all paths

$$S^s(p, d) = \sum_r L_r^s(p, d). \quad (27)$$

The energy function of the entire image is the sum of all the superpixels

$$E(D) = \sum_s S^s(p, d). \quad (28)$$

Consequently, the disparity map can be obtained by minimizing the energy function of the entire image

$$D = \arg\min E(D). \quad (29)$$

The DPSM algorithm is summarized as Algorithm 2.

IV. Experiments

In this section, qualitative and quantitative experiments are carried out on Pleiades and WorldView-2 satellite panchromatic images. Pleiades is composed of two earth observation satellites with the same parameters, which can provide images with a GSD of 0.70 m. In addition, the average GSD of satellite images from WorldView-2 is 0.5 m. The experiments are performed on a personal computer (PC) equipped with Intel Core i7-4790 (3.60 GHz) CPU and 32G RAM. We do not use any parallel programs, GPU, or other dedicated hardware.

The segmentation results from SLIC on BSD which contains 500 (481×321 or 321×481) images have shown that when the number of superpixels was 500, i.e., the average size

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Algorithm 2 DPSM

Input: Stereo-rectified left image I_L, stereo-rectified right image I_R
Output: Disparity map D

/* *Initialization* */
Get superpixel label set S_{set} on I_r by SLIC;
Get matching point set $P_{down}^{l\leftrightarrow r}$ between I_L and I_R by downsampling matching;
Initialize homography model H_i of superpixel i by model fitting;
Initialize joint weighted matching cost C;
/* *The External Propagation* */
Set $\kappa = Area_{superpixel}/2$ to be adaptive to area of original superpixel;
while *the minimum* $Area(i) < \kappa$, $i \in S_{set}$ **do**
  for *each superpixel* j *in the neighborhood of* i **do**
    set $m_{j\to i}(x_i) = \sum_{x_i} \phi_i(x_i)\psi_{j\to i}(x_i, x_j)\prod_{k\in\mathcal{N}(j)/i} m_{k\to j}(x_j)$ (Eq. 20);
    set $b_i(x_i) = \lambda\phi_i(x_i)\prod_{j\in\mathcal{N}(i)} m_{j\to i}(x_i)$ (Eq. 18);
  Compute the label of max belief $L(j_{max}) = \arg\max_{j\in\mathcal{N}(i)} b(x_i)$ (Eq. 24);
  Set $L(i) = L(j_{max})$;
  Update the model of new superpixel j_{max};
  /* *The Internal Propagation* */
  for *each pixel* p *in the new superpixel* j_{max} **do**
    set $L_r^{j_{max}}(p,d) = C^{j_{max}}(p,d) + \min_{q\in\mathcal{N}^r(p)}\left(L_r^{j_{max}}\left(p-r, D_q\right) + V^{j_{max}}\left(D_p, D_q\right)\right)$ (Eq. 26);
    set $S^{j_{max}}(p,d) = \sum_r L_r^{j_{max}}(p,d)$ (Eq. 27);
Set $E(D) = \sum_i S^i(p,d)$ (Eq. 28);
Set $D = \arg\min E(D)$ (Eq. 29);
return D;

TABLE I
PARAMETER SETTINGS FOR THE EXPERIMENTS

Parameters	Description	Value
W_C	Window size of the Census transform	5.0
W_p	Window size of the patch cost	5.0
Thr	Threshold of downsampling matching	0.8
T_d	Downsampling matching factor	4.0
κ	Threshold of regional area in external propagation	162.0
p_1	Parameter of penalty function in internal propagation	0.5
p_2	Parameter of penalty function in internal propagation	3.0
p_3	Parameter of penalty function in internal propagation	2.5
α	The weight of patch cost in joint weighted matching cost	0.3
β	The weight of Census cost in joint weighted matching cost	0.2
γ	The weight of PHOW cost in joint weighted matching cost	0.5

of superpixels was 18 (pixel), both boundary recall (BR) and achievable segmentation accuracy (ASA) of superpixels reached the best. Thus, the size of SLIC superpixel is set to 18 (pixel) in the proposed method, and the other parameters used for the proposed algorithm are listed in Table I.

A. Evaluation of Epipolar Images

Stereo rectification plays a fundamental role in 3-D reconstruction, and two test areas of Pleiades and WorldView-2 images are selected to verify the accuracy of the Affine-compensated RPC stereo rectification [20]. In this article, matching points are employed as check points to calculate errors of stereo rectification. More specifically, the distances of the rectified matching points' Y coordinates on two views are utilized as the errors of stereo rectification. The maximum errors and the Root Mean Square Errors (RMSE) of check points are demonstrated in Table II. The formula of RMSE calculated in this article is as follows:

$$\mathrm{RMSE}\left(x_i^l, y_i^l, y_i^r, H\right) = \sqrt{\frac{1}{m}\sum_{i=1}\left(H\left(x_i^l, y_i^l\right) - y_i^r\right)^2} \quad (30)$$

where (x_i^l, y_i^l) is the coordinate of check point i in the left view, y_i^r is Y coordinates of the matching point i corresponding to (x_i^l, y_i^l) in the right view, $H(\cdot)$ is the affine-compensated RPC stereo rectification.

According to Table II, all the maximum errors are smaller than 1 (pixel), and all the RMSEs are smaller than 0.6 (pixel). Furthermore, to verify the actual errors of the matching points, 45 pairs of matching points are selected randomly as checkpoints, as shown in Fig. 8. It can be considered that the conjugate matching points are located on their epipolar lines or very near to it. In summary, the affine-compensated RPC stereo rectification is able to provide precise results for stereo matching.

TABLE II
ACCURACY OF TEST AREAS FROM PLEIADES AND WORLDVIEW-2

	Max Error (pixel)	RMSE (pixel)
Pleiades		
The 1st Test Area	0.997	0.550
The 2nd Test Area	0.998	0.513
WorldView-2		
The 1st Test Area	0.909	0.510
The 2nd Test Area	0.927	0.560

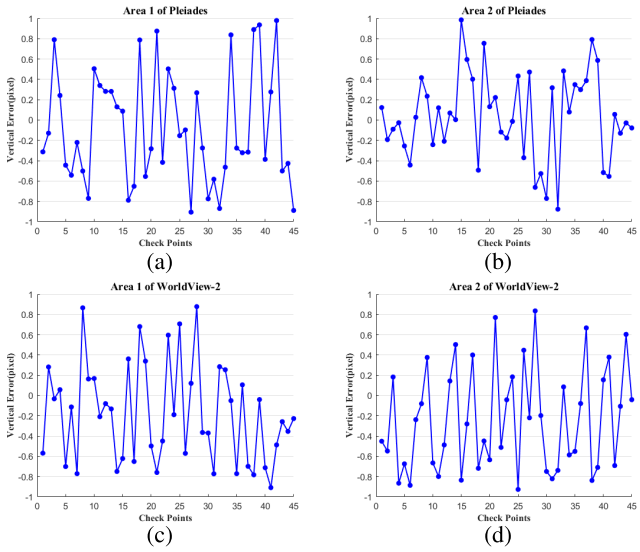

Fig. 8. Distances of rectified matching points' Y coordinates on Pleiades and WorldView-2 satellite imageries. (a) and (b) Scatters of check points in two test areas from Pleiades. (c) and (d) scatters of check points in two test areas from WorldView-2.

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TABLE III

NUMBERS OF MATCHING POINTS FROM THE FIVE ALGORITHMS ON PLEIADES AND WORLDVIEW-2 SATELLITE IMAGES

	SIFT	ASIFT	LLT	FD	**Proposed**
Pleiades	486	7741	461	545	**23309**
WorldView-2	68	789	53	31	**13800**

TABLE IV

NUMBER OF MATCHING POINTS CONTAINED IN PER 100 PIXELS FROM THE FIVE ALGORITHMS ON PLEIADES AND WORLDVIEW-2 SATELLITE IMAGES

	SIFT	ASIFT	LLT	FD	**Proposed**
Pleiades	0.130	2.076	0.124	0.150	**6.250**
WorldView-2	0.031	0.356	0.024	0.014	**6.250**

B. Evaluation of Downsampling Matching

In order to verify the effectiveness of the proposed downsampling matching, we compared the proposed approach with four state-of-the-art feature matching methods, including SIFT [35], affine SIFT (ASIFT) [40], locally linear transforming (LLT) [41] and fast dense feature-matching (FD) [42], and the parameters are selected recommended by their suggestions. Moreover, different subimages from Pleiades and WorldView-2 satellite are selected as test areas, whose size are 572×652 and 552×401, respectively.

The results of the comparison experiments are shown in Fig. 9. It can be clearly found that matching points of SIFT, ASIFT, LLT and FD are very uneven and mainly distribute on the edges of the buildings and roads with rich texture, there are few matching points on the roofs of buildings and surfaces of roads with weak texture. On the contrary, the proposed downsampling matching can provide uniform-distribution matching points on the entire images with different objects. According to the number of matching points about the five algorithms, as shown in Table III, the number of proposed matching points is the largest, which is due to the controllable downsampling density. Additionally, the distribution of downsampling matching points is robust to the different environments, as shown in Table IV. On the contrary, SIFT, ASIFT, LLT and FD are very sensitive to image scenes by observing the second row of Table IV. The precision of the proposed downsampling matching is not the best, as shown in Table V, it is acceptable since that model fitting based on RANSAC is good at filtering false matching points. The calculation of the precision about the five matching algorithms is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{31}$$

where TP is the number of correct matching points and FP is the number of wrong matching points.

In summary, the proposed downsampling matching is appropriate for model fitting on different satellite images.

C. Evaluation of Three Matching Costs

In this section, the effectiveness of patch cost, census cost, and PHOW cost are verified qualitatively on a subregion of satellite images. Disparity maps from three different matching costs optimized individually by the proposed double propagation are shown in Fig. 10.

As seen from Fig. 10, many errors appear on disparity maps when the three matching costs are used alone. Although patch cost is robust to weak and similar textures, it also results in some extra buffers of edges leading to extended edges [e.g., Fig. 10(b)] as mentioned in *II. Related Work*. Obviously,

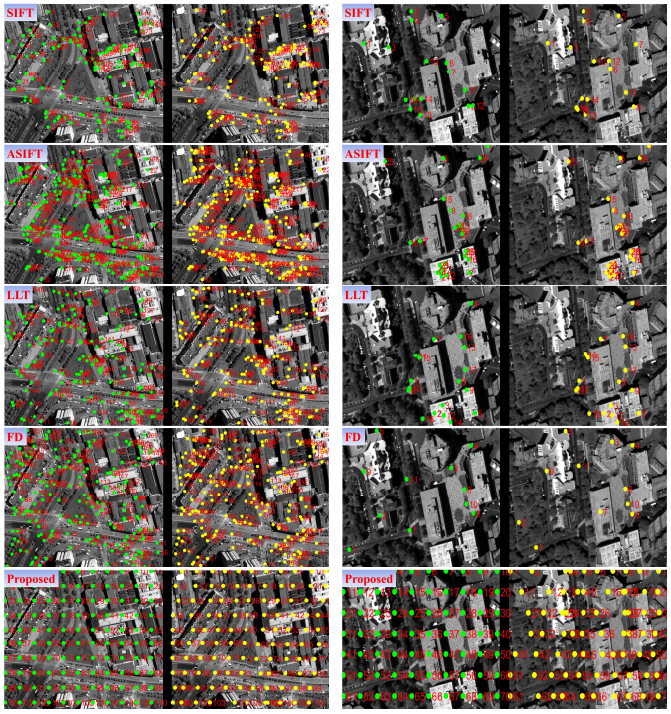

Fig. 9. Match point distribution of SIFT [35], ASIFT [40], LLT [41], FD [42] and proposed downsampling matching. (a) Results of Pleiades satellite images. (b) Results of WorldView-2 satellite images. From top to bottom: The results of SIFT (shown with 3 times sparseness), the results of ASIFT (shown with 30 times sparseness), the results of LLT (shown with 3 times sparseness), the results of FD (shown with 3 times sparseness) and the results of the proposed downsampling matching (shown with 225 times sparseness). The matching points in different views are represented by dots with different colors, and the corresponding points are marked with the same numerical labels.

TABLE V

PRECISION OF MATCHING POINTS FROM THE FIVE ALGORITHMS ON PLEIADES AND WORLDVIEW-2 SATELLITE IMAGES

	SIFT	ASIFT	LLT	FD	**Proposed**
Pleiades(%)	92.49	98.08	97.40	98.38	**95.45**
WorldView-2(%)	64.58	98.75	96.30	80.65	**85.71**

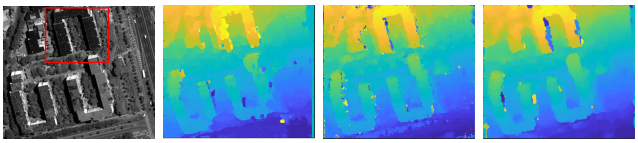

Fig. 10. Disparity maps from three different matching costs individually. (a) Left view of stereo pair. (b) Disparity map of patch cost. (c) Disparity map of Census cost. (d) Disparity map of PHOW cost.

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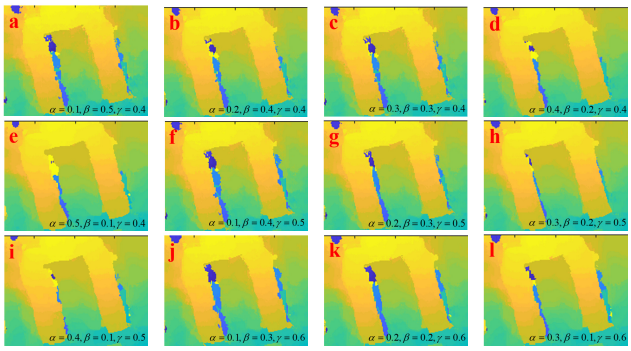

Fig. 11. Stereo matching results from three matching costs with different weight parameters. (a)–(e) Disparity maps when $\gamma = 0.4$. (f)–(i) Disparity maps when $\gamma = 0.5$. (j)–(l) Disparity maps when $\gamma = 0.6$.

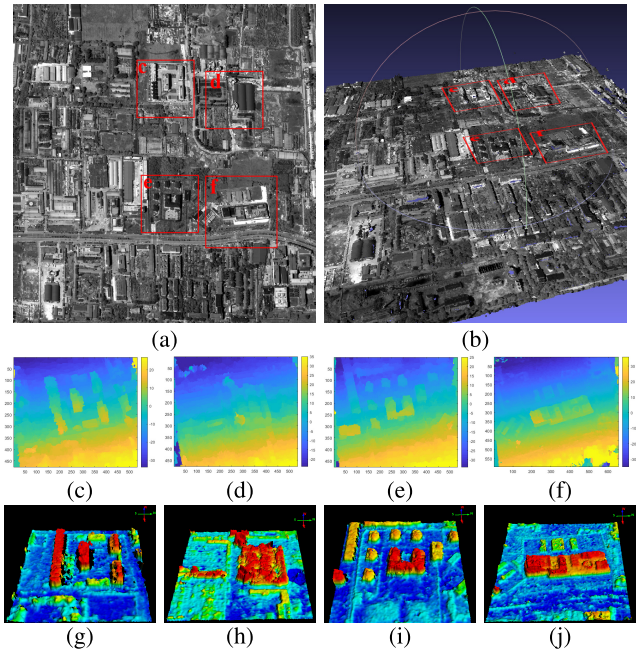

Fig. 12. Reconstruction results of Pleiades satellite images. (a) Left view of stereo pair. (b) Unrendered 3-D point cloud corresponding to (a). (c)–(f) Disparity maps corresponding to the labels in (a). (g)–(j) Enlarged DSMs corresponding to the labels in (a).

census cost is able to acquire parallaxes in both shaded and illuminated areas, but it leads to some ambiguities between target pixels [e.g., Fig. 10(c)]. In addition, even though PHOW causes some errors, the disparity map of which is better than patch cost and Census cost generally. Given these results, we designed experiments to verify joint weighted matching cost under the condition of different weight parameters, and α, β and γ represent the weight of patch cost, census cost, and PHOW cost, respectively. According to the independent matching results of three costs, the weight of PHOW cost γ is set to 0.4, 0.5, and 0.6 in these experiments. Furthermore, we control the value of γ to observe the influence of changes in α and β on the experimental results, and keep the condition of $\alpha+\beta+\gamma = 1$. The results of stereo matching with different parameters on a subregion of Fig. 10(a) are shown in Fig. 11.

As seen from Fig. 11, when the value of γ is fixed, the bigger value of α is, the larger proportion of building edges

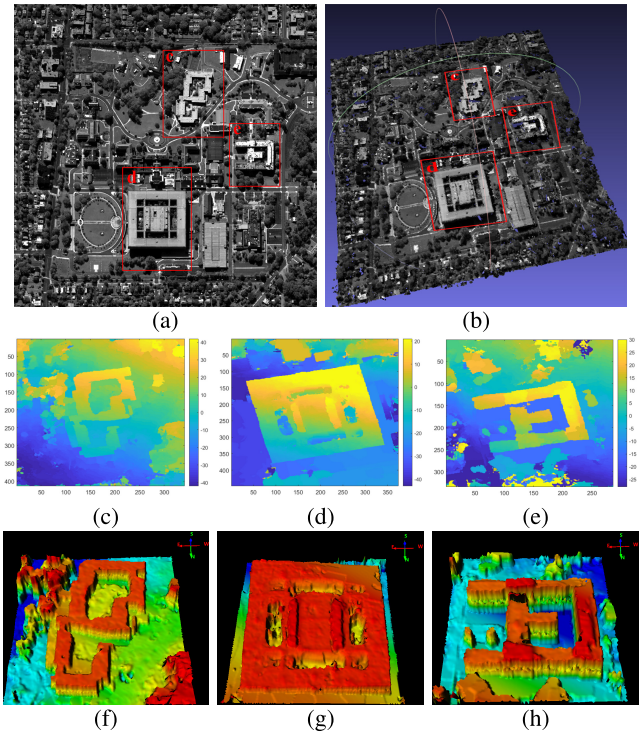

Fig. 13. Reconstruction results of WorldView-2 satellite images. (a) Left view of stereo pair. (b) Unrendered 3-D point cloud corresponding to (a). (c)–(e) Enlarged disparity maps corresponding to the labels in (a). (f)–(h) Enlarged DSMs corresponding to the labels in (a).

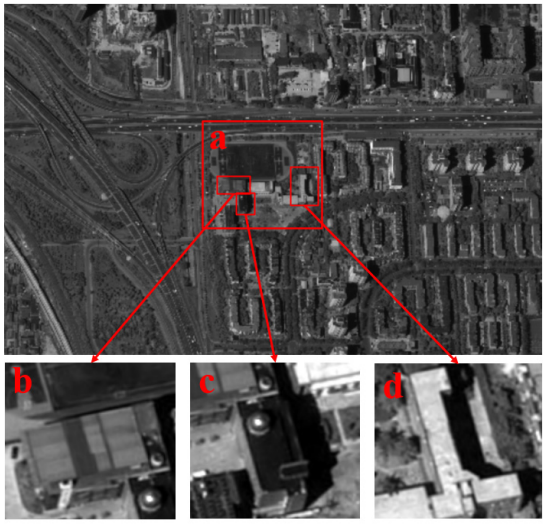

Fig. 14. Left rectified Pleiades satellite imagery for qualitative comparison experiments. (a) Subregion of left view. (b)–(d) Enlarged subareas corresponding to the red boxes in (a).

is. At the same time, the accuracy of the edges can be adjusted by changing the value of γ. By comprehensively observing the impact of the three parameters on the stereo matching results, we find that when $\alpha = 0.3$, $\beta = 0.2$ and $\gamma = 0.5$, both the accuracy of the building edges and the smoothness of the flat areas reached the best [e.g., Fig. 11(h)]. Therefore, we will set the values of α, β and γ to 0.3, 0.2, and 0.5 in the following comparison experiments.

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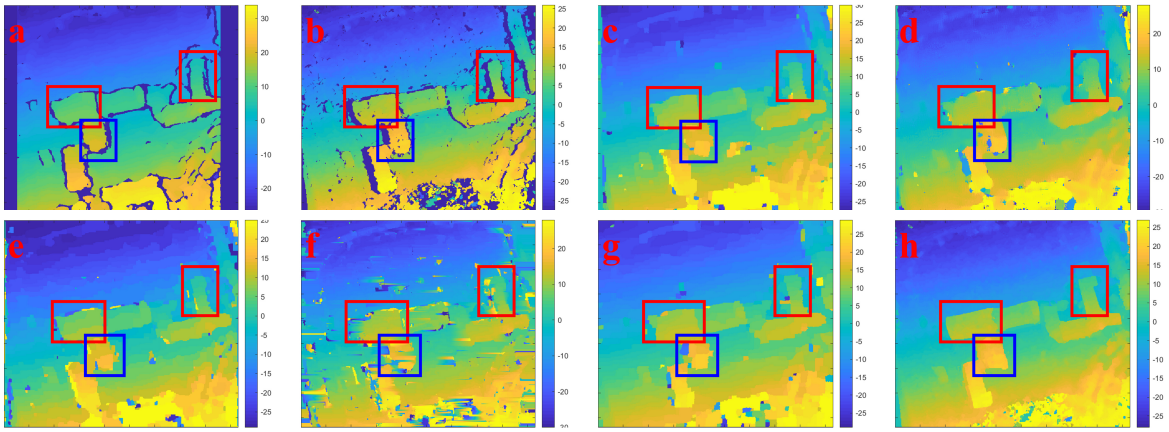

Fig. 15. Disparity maps of different stereo matching algorithms. (a) SGM [11]. (b) MGM [30]. (c) PMBP [12]. (d) SDRL [21]. (e) ImpCensus-GC [13]. (f) SGBM [23]. (g) ICC-ASGF [14]. (h) Proposed algorithm. The boxes correspond to Fig. 14(b)–(d).

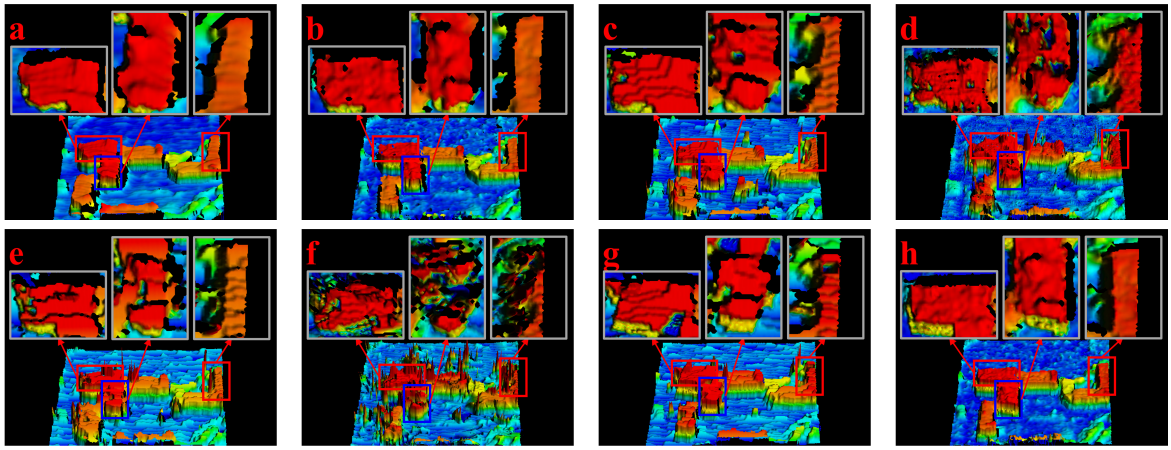

Fig. 16. DSMs of different stereo matching algorithms. (a) SGM [11]. (b) MGM [30]. (c) PMBP [12]. (d) SDRL [21]. (e) ImpCensus-GC [13]. (f) SGBM [23]. (g) ICC-ASGF [14]. (h) Proposed algorithm. The boxes correspond to Fig. 14(b)–(d). The viewing angles of all DSMs are from east to west.

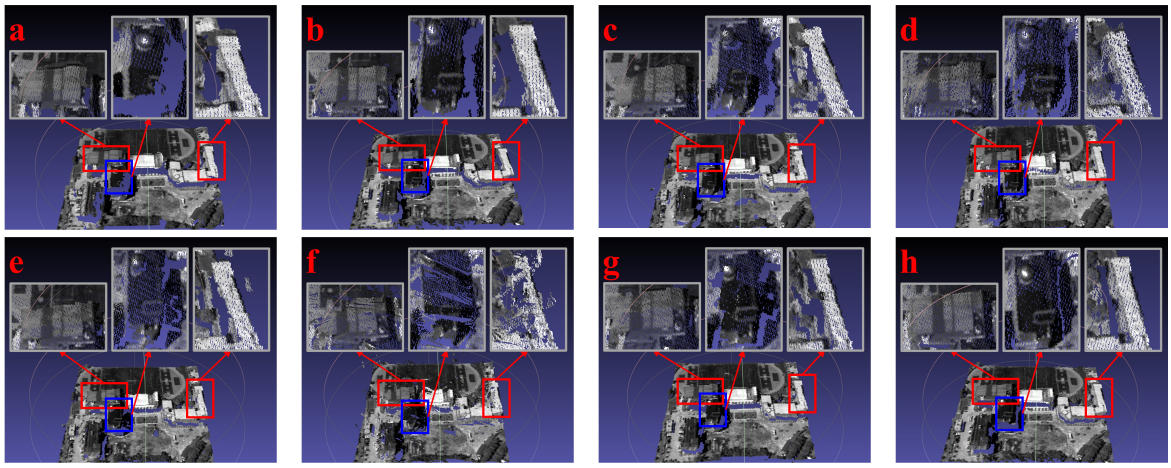

Fig. 17. 3-D point clouds of different stereo matching algorithms. (a) SGM [11]. (b) MGM [30]. (c) PMBP [12]. (d) SDRL [21]. (e) ImpCensus-GC [13]. (f) SGBM [23]. (g) ICC-ASGF [14]. (h) Proposed algorithm. The boxes correspond to Fig. 14(b)–(d). The viewing angles of all 3-D point clouds are from east to west.

D. Evaluation of DPSM

In this section, the effectiveness of the proposed stereo matching algorithm is verified qualitatively by experiments on different satellite images. Giving that the proposed method is employed to 3-D reconstruction on satellite images, the DSMs and 3-D point clouds are served as evaluation results.

The first pair of test images are Pleiades stereo imageries from Beijing, China, the results of which are shown in Fig. 12, and Fig. 12(a) is the left imagery with the size of 2500×2500. The qualitative evaluation results show that the proposed algorithm is able to calculate accurate disparity maps of satellite images, then DSMs and 3-D point clouds can be reconstructed reliably based on our disparity results. More narrowly, it can be

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TABLE VI
ANALYSIS OF ACCURACY AND TIME EFFICIENCY ABOUT THE EIGHT ALGORITHMS

	SGM	MGM	PMBP	SDRL	ImpCensus-GC	SGBM	ICC-ASGF	**Proposed**
Error type								
RMSE (m)	3.08	4.30	4.42	4.23	6.80	4.52	2.36	**2.07**
Max error (m)	9.37	27.34	9.85	10.49	12.60	8.90	5.31	**3.14**
Speed								
Time (min)	4.18	0.29	19.87	18.20	16.53	10.73	23.80	**7.40**

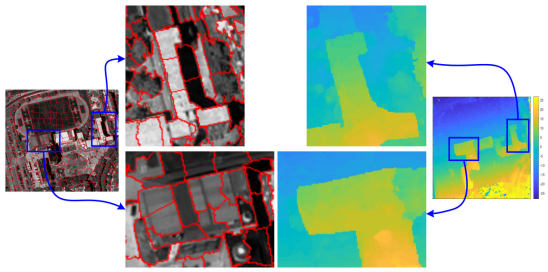

Fig. 18. Correspondences between superpixel boundaries and disparity maps. Left to right: The superpixel result, the closeups of superpixel result, the closeups of disparity map, the disparity map.

also found that different disparity faults are accurately *located* at the intersection between buildings and ground, the satisfactory disparity results of weak texture regions are also obtained [e.g., Fig. 12(c) and (e)]. Furthermore, the calculated DSMs reflect the real 3-D structure of the buildings in urban scenes [e.g., Fig. 12(g)–(j)].

However, when it comes to the edges of disparity maps, some errors come up, which is caused by the fact that the left and right stereo images contain different ground objects in the edge areas. Moreover, an affine matrix is used to complete a rotation compensation on the whole cutting image in the stage of stereo rectification. Thus, the disparity values gradually increase from the upper left corner to the lower right corner in each disparity map.

The second pair of test images are Worldview-2 stereo images from Washington dc, USA, the results which are shown in Fig. 13, and Fig. 13(a) is the left imagery with the size of 1800 × 1800. Generally, the DSMs of the sharp corners of buildings as well as the depth fault areas between roofs and ground are fine [e.g., Fig. 13(f) and (g)]. Unfortunately, comparing with Pleiades, the DSMs and 3-D point clouds of WorldView-2 have more holes, especially the edges of reconstruction buildings are noisy [e.g., Fig. 13(b), (g) and (h)]. This is due to the fact that the photographic field of WorldView-2 is so complex that numerous shadows and trees are included. Additionally, the viewing angle of WorldView-2 is greater than Pleiades, so severely distorted areas may appear in the direction of deviating from the satellite shooting angle.

E. Comparison With State-of-the-Art

1) Qualitative Comparison Experiments: To demonstrate the effectiveness and time efficiency of the proposed algorithm comprehensively, some state-of-the-art algorithms that are in common with the proposed algorithm are selected for contrast experiments in this section. The similarities are summarized as follows:

1) The optimization of dynamic programming is employed in SGM [11] and MGM [30], which is similar with the internal propagation of the proposed algorithm.
2) The support-window-based matching cost is employed in patch match belief propagation (PMBP) [12], imperfectly Census and graph-cut matching (ImpCensus-GC) [13] and ICC-ASGF [14], which is similar with the Patch term and Census term of joint weighted matching cost in the proposed algorithm.
3) The SIFT-feature-based matching cost is employed in sub-pixel disparity refinement based on Lagrange interpolation (SDRL) [21], which is similar to the PHOW term of joint weighted matching cost in the proposed algorithm.
4) The edge-block matching is employed in semi-global and block matching (SGBM) [23], which is similar to the constraints of region boundaries in the proposed algorithm.

Therefore, the comparison algorithms in this section include: SGM [11], MGM [30], PMBP [12], SDRL [21], ImpCensus-GC [13], SGBM [23] and ICC-ASGF [14] (sorted by time). The parameters of all comparison methods are set following their articles' suggestion. Additionally, we compared the DSMs and point clouds from all methods using same parameters in the step of triangulation. For concreteness, the left rectified Pleiades satellite imagery with the size of 465 × 419 is shown in Fig. 14(a), which contains buildings, roads and trees. The reconstruction results of these eight algorithms corresponding to Fig. 14(a) are shown in Figs. 15–17.

According to Figs. 15 and 16, the generated disparity maps and DSMs of all the eight algorithms can clearly describe the most ground, buildings, and streets in urban scenes, but not SGBM. There seems to be no effective means to eliminate the errors of edge-block matching and these errors are enlarged by linear interpolation in SGBM. Therefore, plenty of strip-type errors appear on the disparity map of SGBM [e.g., Fig. 15(f)]. Conversely, the disparity map obtained by the proposed algorithm keeps ahead of SGBM [e.g., Fig. 15(h)], since that initial segmentation results were iteratively optimized by the external propagation in the proposed algorithm.

Moreover, when it comes to the parallax faults between buildings and ground as well as the occlusions of buildings, SGM and MGM could only obtain rough disparity maps [e.g., the red boxes in Fig. 15(a) and (b)]. Thereby, there are many errors in DSMs and point clouds too [e.g., the red boxes in Figs. 16(a) and (b) and 17(a) and (b)]. It is difficult for dynamic programming methods to compute edge-pixel parallaxes only by the optimization of cost aggregation.

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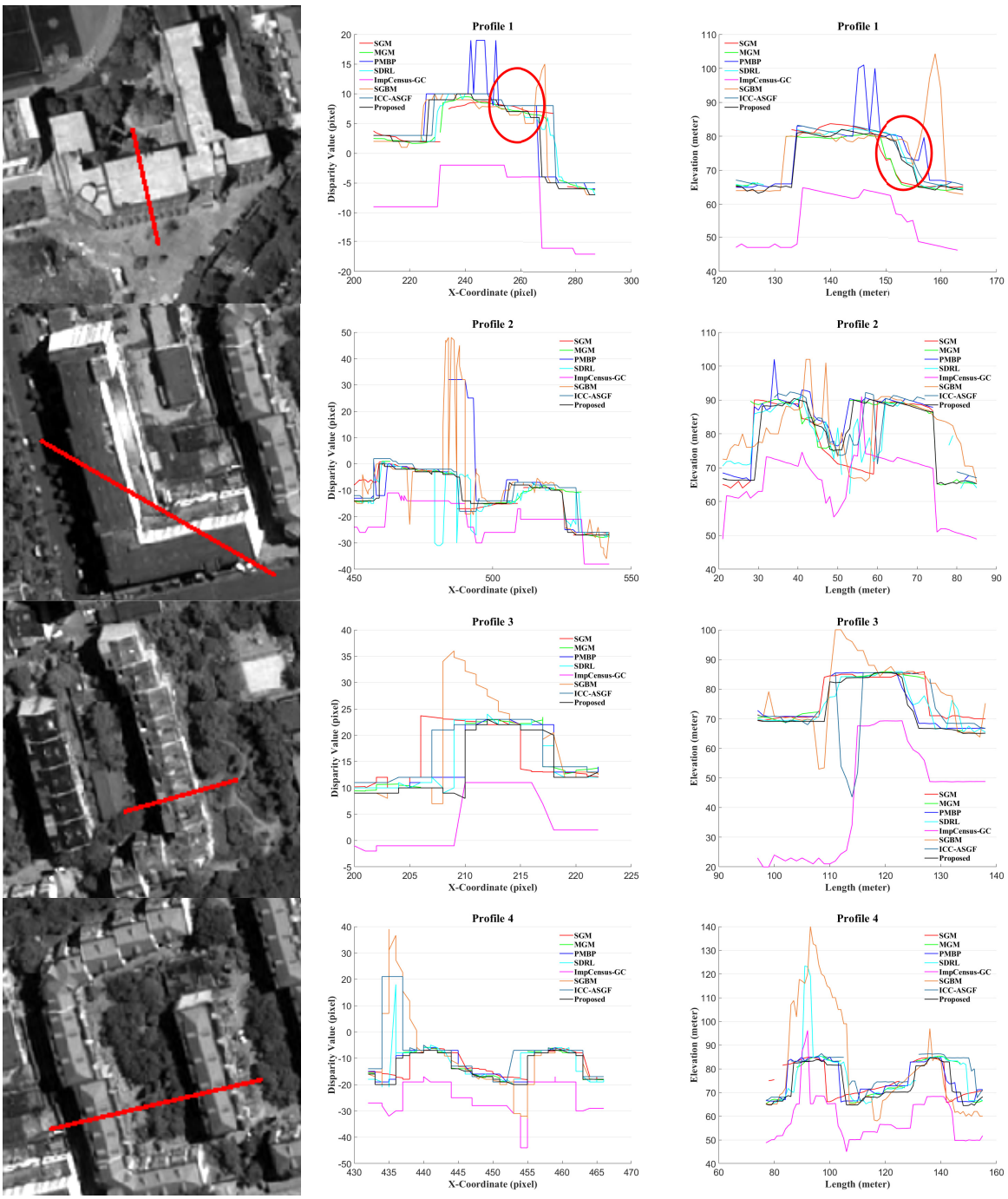

Fig. 19. Profiles of disparity maps and DSMs of the eight algorithms. (a) Four subareas of the left view, the red line segments represent profiles. (b) Disparity scattered diagrams corresponding to four profiles in (a), horizontal axis and vertical axis denote the length (pixel) along the path and disparity value (pixel), respectively. (c) Elevation scattered diagrams corresponding to four profiles in (a), horizontal axis and vertical axis denote the length (meter) along the path and elevation value (meter), respectively.

Furthermore, disparity values of missing pixels at occlusions have little prospect of being worked out. On the country, the parallaxes of depth discontinuous areas can be optimized by internal propagation under the guidance of the region edges, the punishment of cost on edges permits parallax jumps, as seen from the correspondences between superpixel boundaries and disparity maps (e.g., Fig. 18). Moreover, the parallaxes of partial occlusions are able to be inferred out under the constraints of superpixel edges in the proposed method [e.g., the red boxes in Fig. 17(h)].

There are many extended edges and numerous square-shaped errors that come from support-windows on the disparity maps of PMBP, ImpCensus-GC, and ICC-ASGF [e.g., Fig. 15(c), (e) and (g)]. We also found that many spotted

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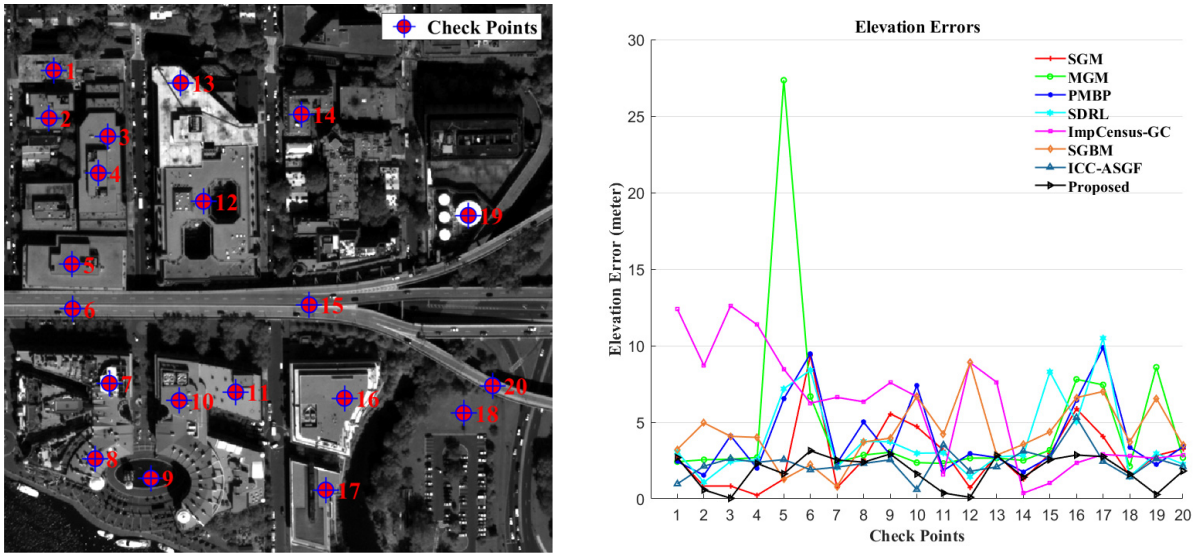

(a) (b)

Fig. 20. Reconstruction area and check point distribution map. (a) Distribution map of 20 check points on WorldView-2 satellite image. (b) Scattered diagram of elevation errors corresponding to (a).

errors appear on Fig. 15(d), these errors appear as spikes in DSM from SDRL [e.g., Fig. 16(d)]. The reason is that feature matching lacks an effective smoothing strategy, resulting in too many outliers in the disparity map that cannot be used for accurate 3-D reconstruction.

Unfortunately, objects in the shaded areas are hardly visible, which barriers to consistent computing in all of these algorithms, including the proposed algorithm [e.g., the blue boxes in Fig. 15(h)], so DSMs and point clouds also have some outliers [e.g., the blue boxes in Figs. 16(h) and 17(h)]. Binocular stereo vision relies strongly on the visible objects, which puts forward higher requirements for the acquisition of satellite images.

In order to evaluate the results of eight algorithms more intuitively, we conduct a building profile analysis on the disparity maps and DSMs. Y-axis values are acquired from Z values of the discrete points along the profiles and the X-axis values are their distances along the profiles in the scattered diagrams, as shown in Fig. 19. Benefited from optimization of double propagation, the profiles of buildings are rebuilt well through the proposed method, as seen from the black lines in Fig. 19(b) and (c). Furthermore, the profile 1 is located on an inclined surface of a building [e.g., the first row of Fig. 19(a)], the proposed algorithm displays obvious advantages compared with the other seven algorithms [e.g., the red circled regions in the first row of Fig. 19(b) and (c)]. The results benefit from the homography models of inclined surfaces in the external propagation. Additionally, numerous disparity values and elevations of the other algorithms are wrong or missing in-depth discontinuous areas where the roofs and the facades meet. In particular, the stair-stepping errors appear on the profiles from SGBM, as shown in profile 3 and profile 4. This phenomenon also validates the errors of linear interpolation in SGBM we mentioned above.

According to Fig. 19(b) and (c), the curves of disparity do not exactly match the curves of elevation, we own this

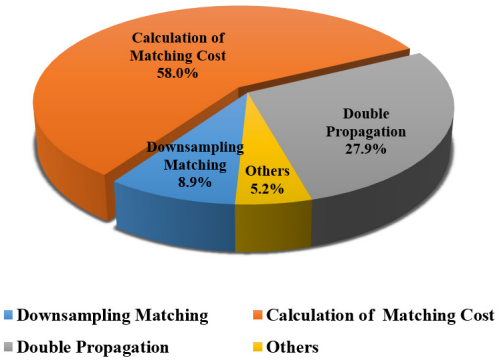

Fig. 21. Time distribution of each part in the proposed algorithm.

difference to the errors of triangulation, and the inaccurate stereo correction matrix would also cause the collection points of disparity maps and DSMs to not be in the same positions.

2) Quantitative Comparison Experiments: Until now, the lack of suitable calibrated satellite images with known ground truth disparity maps has prevented such direct comparisons. Therefore, the WorldView-2 satellite image with LiDAR ground truth DSM is used to evaluate the accuracy of eight algorithms in this section. To evaluate the accuracy of eight algorithms, 20 check points distributed on buildings, roads and ground are collected on a test area of satellite image, as shown in Fig. 20(a). The error distribution of check points about the eight algorithms is shown in Fig. 20(b). We can find that MGM has a maximum error point and ImpCensus-GC is most sensitive to changes in scenes, on the contrary, the proposed method is the most robust to scene changes among the eight algorithms. In particular, compared with other areas, all of these eight algorithms have larger elevation errors in weak-texture areas [e.g., the check point 9, 16, and 17 in Fig. 20(b)], which was failed to be reflected in the qualitative comparison experiments.

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In addition, the root mean squared error (RMSE), max error, and actual running time on the test area with the size of 700×700 are utilized to analysis the accuracy and time efficiency of these eight algorithms, the results are presented in Table VI. Observe that not only RMSE but also max error of the proposed algorithm is the lowest among the eight algorithms. In fact, the calculation of matching cost is the most time-consuming by analyzing the running time of each part in the proposed algorithm (e.g., Fig. 21). Nevertheless, the calculation of matching cost also can be sped up through parallel processing. Besides, the MGM algorithm is evaluated by a compiled program, so the actual running time of which is the least. Based on the comparative experiments, the proposed algorithm is more suitable for 3-D reconstruction on urban satellite remote sensing images than the others.

V. Conclusion and Future Works

To improve the effectiveness of 3-D reconstruction from urban satellite images, we have proposed the DPSM method which starts with initialization of regional models consisting of superpixel segmentation, downsampling matching, and model fitting. As an extension of the conventional matching cost, the joint weighted matching cost is utilized to obtain the credible global similarity metric. Subsequently, a novel technique is developed, namely double propagation, which formulates the disparity refinement problem as an information propagation by optimizing iteratively regional models and region boundaries in external propagation. Moreover, the matching cost of each pixel is refined through penalty propagation along the scanning lines under the restrictions of region boundaries in internal propagation.

Evaluations on satellite images from Pleiades and WorldView-2 demonstrate that the proposed method not only can preserve the quality of DSMs and 3-D point clouds at depth faults (e.g., building edges), but also restore the planarity for piecewise structures in slant areas (e.g., the roofs of buildings). Especially, parallaxes of partially occlusions are able to be inferred based on regional models.

However, when applying this method to rural or forest areas, the problem becomes tough, as it is difficult for geometric features and superpixel edges proposed in this article to perform effectively in these areas with irregular edges and characteristics of seasonal differences. It is also a challenge for two-view stereo methods, including DPSM, to acquire accurate parallaxes in areas with obvious temporal differences, including vegetations and shadows, even though we introduced additional geometric information as an assistant. To replenish enough color information in these areas, multiview stereo tries to provide redundant image information from multiple perspectives. Unfortunately, this also increases additional equipment cost to capture more satellite images. And in particular, we find that both shadows and occlusions are related to facades which block the ground and cause shadows. How to identify the facades and make use of its vertical relationship with the ground to restore the occlusions and shadow areas may be a new research field in 3-D reconstruction. Furthermore, the disparity maps of the proposed algorithm are dependent on the superpixel results, and the precise semantic segmentation results are beneficial for our method empirically. The 3-D reconstruction with semantic information is also a promising research hotspot.

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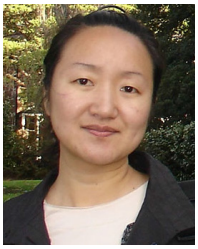
His research interests include computer vision, image processing and segmentation, stereo vision, and 3-D reconstruction.



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